

Can a Connection Remember?

DataForge 2026 — Explain the Frontier | Synaptic Plasticity as Short-Term Memory

Purpose. This blog explains the mechanism behind our interactive learning artifact from first principles, then connects the simplified substrate to recent research and the Dragon Hatchling family. The project is intentionally an educational toy rather than a biological simulator.

1. The question

The artifact starts with one question: **Can a connection remember?** A learner teaches an association, watches its synaptic state change, waits while that state fades, and introduces newer information. The learner then predicts what the model will recall. This makes the concept something the learner can manipulate rather than merely read about.

2. Neurons, synapses, and plasticity

A neuron receives, processes, and sends signals. Neurons communicate at synapses. Synaptic plasticity means that synaptic effectiveness can change. Short-term plasticity refers to temporary changes that can later fade. Garcia Rodriguez et al. (2022) describe STP as a mechanism for decaying memories in synapses and introduce STP Neurons whose synapses carry state through time [1].

3. What is a weight?

In our toy, a **weight is simply a number representing the current strength/state of an association**. If Apple → Red has weight 0.8 and Apple → Green has weight 0.2, Red is currently stronger. The weight is not a probability and not a biological measurement. It is the numerical state that gives the toy a place to store the effect of previous learning.

4. What is decay?

Decay is the gradual fading of the stored state. If a weight starts at 1.0 and the decay factor is 0.8, passive steps give 0.8, then 0.64, then 0.512. In general, after n passive steps the toy gives $\delta^n w(t)$. A higher δ means more of the old state survives each step. We call this persistence in the interface so the learner can connect the parameter to a memory trace that hangs around.

5. What is plasticity or learning rate?

The learning rate η controls how strongly a new learning event changes the state. The simplified rule is **$w(t+1) = \delta w(t) + \eta \cdot \text{pre} \cdot \text{post}$** . The first term preserves part of the old state; the second adds a new learning contribution when the input and target are active. This is a pedagogical Hebbian-style rule, not a complete biological model.

6. From equation to experiment

The learner first teaches Apple → Red repeatedly. Red grows. Time passes, so Red decays. Recall chooses the strongest current association. Then Apple → Green is taught once. Red and Green remain separate traces in this implementation. They compete at readout because the model compares their current weights and selects the maximum. The toy does not claim that Green physically overwrites Red.

7. The stability–plasticity trade-off

A useful memory system needs to respond to new information without instantly losing everything it learned a moment ago. Higher persistence favors retention of older traces. Stronger plasticity favors rapid response to new events. These pressures can conflict. Our final challenge lets the learner vary persistence/decay and delay and test how the outcome changes.

8. Research evidence

Kozachkov et al. (2022) trained recurrent networks with and without STSP on a working-memory task. They reported that both could maintain memories despite distractors, while STSP networks showed more brain-like activity and greater robustness to network degradation [2]. This is evidence from computational models, not proof that the human brain uses our exact rule.

Chrysanthidis et al. (2025) built a spiking network model to study episodic memory and short-term recency effects through synaptic mechanisms across timescales [3]. Our Apple/Red/Green experiment is an educational simplification inspired by that broader question, not a reproduction of the published model.

9. Why BDH belongs in the story

Kosowski et al. (2025) describe BDH (The Dragon Hatchling) as a brain-inspired language-model architecture whose working memory during inference relies on synaptic plasticity with Hebbian learning using spiking neurons [4]. That makes BDH a substantive technical connection: changing synaptic state is part of a frontier architecture. Our toy isolates the intuition in a tiny transparent substrate; it is not BDH.

BDH-CQ (Engdahl et al., 2026) extends the BDH family toward in-context learning with recurrent latent reasoning [5]. Demonstrations update an evolving recurrent memory and the model then performs latent computation. Our project does not implement this mechanism. We mention BDH-CQ to place the selected concept in the current BDH research lineage without inventing a direct correspondence.

10. What our artifact actually computes

The computational core lives in `model.py`. ``learn()`` applies the update rule; ``step_time()`` applies decay; ``recall()`` selects the maximum-weight label; and ``simulate_scenario()`` computes the final challenge from selected conditions. The app uses those states for numerical displays, curves, and the synapse visualization. There are no precomputed challenge outcomes.

11. What is simplified

Real synaptic plasticity involves many biological mechanisms, timescales, and state variables. Our toy uses one scalar weight per association, a simplified update rule, multiplicative decay, and argmax readout. It does not simulate spikes, calcium dynamics, vesicle release, STDP, attractor dynamics, or the high-dimensional states of frontier architectures. The visualization is a teaching representation: geometry and particle flow communicate the numerical state rather than literal biological geometry.

12. Why a toy model?

The DataForge track requires an explorable learning artifact. A tiny transparent model makes the mechanism inspectable: the learner can see the number, change conditions, make a prediction, and immediately test it. The goal is not to simulate every detail of the brain; it is to make one precise technical claim reproducible in under a minute.

13. Recommended demo

Start from the clean baseline. Teach Apple → Red. Point to the weight and say that it is the current numerical synaptic state. Let time pass and explain decay. Recall. Teach Apple → Green and ask for a prediction. Resolve the readout. Finish with the challenge: change persistence and delay, predict, run, and compare the prediction with the computed intermediate values.

14. Honest takeaway

The strongest defensible takeaway is narrow: **a changing synaptic state can function as a temporary computational memory trace, and persistence plus new learning can affect what a simple readout retrieves.** Our artifact demonstrates that behavior in its own toy substrate. The cited research shows related mechanisms in richer systems; it does not justify saying that our toy proves a biological theory of human memory.

References

[1] Garcia Rodriguez, H., Guo, Q., & Moraitis, T. (2022). Short-Term Plasticity Neurons Learning to Learn and Forget. PMLR. <https://proceedings.mlr.press/v162/rodriguez22b.html>

[2] Kozachkov, L. et al. (2022). Robust and brain-like working memory through short-term synaptic plasticity. PLOS Computational Biology. <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1010776>

[3] Chrysanthidis, N. et al. (2025). Short-term plasticity influences episodic memory recall. Scientific Reports. <https://www.nature.com/articles/s41598-025-12611-5>

[4] Kosowski, A. et al. (2025). The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. <https://arxiv.org/abs/2509.26507>

[5] Engdahl, B. et al. (2026). BDH-CQ: In-Context Learning with Recurrent Latent Reasoning. <https://arxiv.org/abs/2608.09888>